MineTrace - Critical Mineral Supply-Chain Risk Assessment

REPORT ID: REP-4D4180A12C2F | ANALYSIS ID: LIST-REP-1 | TEMPLATE VERSION: 1.0

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1. Executive Summary

CLASSIFICATION: RED (ANOMALY SCORE: 60.0/100) - COMPLETE
Flag Status: FLAGGED | Sanctions Screening: NO_IDENTIFIED_MATCH | Evaluated Rules: 5 | Insufficient Rules: 0

Mineral / Commodity:	Lithium carbonate (HS 283691)	**Reporter Country:**	CHL
Partner Country:	USA	**Trade Period:**	2023
Score Completeness:	COMPLETE	**Sanctions Screening:**	NO_IDENTIFIED_MATCH

2. Risk Rule Evaluation Details

Rule ID	Rule Name	Weight	Outcome	Threshold	Calculated
R1	Mirror Trade Gap	30 pts	FLAG	> 30.0%	45.0%

[R1] Mirror Trade Gap (Weight: 30 pts) - Status: FLAG
Formula: abs(export_value - import_value) / max(export_value, import_value) * 100 | Threshold: > 30.0% | Calculated: 45.0%
Inputs: {'export_value': 1200000.0, 'import_value': 660000.0}
Explanation: Export-import mirror trade gap 45.0% exceeds threshold.

3. Classification & Sufficiency Guardrails

Saved Anomaly Score:	60.0 / 100.0	**Assigned Tier:**	RED
Flag Status:	FLAGGED	**Policy Version:**	1.0

Thresholds Applied: GREEN (0-19), AMBER (20-49), RED (50-100), INSUFFICIENT_DATA (PARTIAL/UNAVAILABLE)
Classification Rationale: Elevated anomaly score exceeding RED prototype threshold (>= 50.0).

4. Sanctions Screening (OFAC SDN)

SANCTIONS STATUS: NO_IDENTIFIED_MATCH
Registry Reference: OFAC Specially Designated Nationals (SDN) List Limitations:
SANCTIONS SCREENING LIMITATIONS: Screening is performed against public US Treasury OFAC Specially Designated Nationals (SDN) registries. Country-level or commodity-level screening does not substitute for corporate beneficial ownership checks (e.g., 50 Percent Rule) or end-user certification.

5. Source Evidence & Audit Trail

Source Dataset:	UN Comtrade	**Source Record ID:**	TRADE-1
Retrieval Timestamp:	2023-05-01T10:00:00	**Evidence Snapshot ID:**	SNAP-LIST-REP-1
Validation Status:	VALID	**Source Endpoint / URL:**	https://comtradeapi.un.org/public/v1/pre

6. Validated AI Explanation

Status: NOT_REQUESTED. No pre-validated AI explanation was attached to this transaction record.

7. Methodology, Limitations & Legal Notice

NOTICE & DISCLAIMER: This document is an automated evidence report generated by the MineTrace critical mineral supply-chain risk intelligence system. Numerical anomaly scores, rule evaluations, and risk classifications (GREEN, AMBER, RED, INSUFFICIENT_DATA) reflect deterministic statistical and relational anomalies compared against historical baselines and official trade registries. They DO NOT constitute a legal finding of fraud, illicit trade, customs violation, or sanctions evasion. A GREEN classification does not certify a supply chain as compliant, ethical, or risk-free. Missing or incomplete data must never be interpreted as low risk.

METHODOLOGY & LIMITATIONS:

1. Anomaly Score: A deterministic weighted score (0-100) aggregating multi-rule trade indicators.
2. Scope of Detection: Risk rules evaluate reported trade flows (Comtrade) and public SDN lists (OFAC). They do not inspect physical shipping containers or corporate beneficial ownership trees.
3. Data Sufficiency: Where required historical baselines or partner trade reports are unavailable, MineTrace strictly enforces sufficiency guardrails and assigns INSUFFICIENT_DATA rather than assuming compliance.
4. Sanctions Screening: Evaluates exact and normalized name matches against US Treasury OFAC records. A status of NO_IDENTIFIED_MATCH does not verify compliance with non-US regimes or unlisted intermediaries.
5. Prototype Status: Thresholds (0-19 GREEN, 20-49 AMBER, 50-100 RED) are heuristic operational baselines designed for screening and prioritization, not statistically validated probabilities of wrongdoing.